

Can light change what a plant makes?

Learning intention

Describe photosynthesis and investigate how light intensity affects a measured proxy for its rate.

Mission: make a prediction, test the method and defend a conclusion.

A good investigation records a limit as well as a pattern.

Materials, energy and measurement

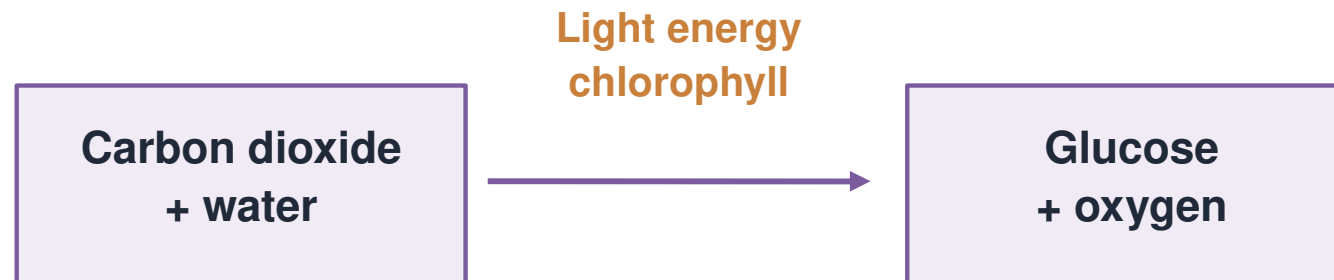
R1. Where is chlorophyll found in plant cells?

R2. What does a control variable do?

R3. What does “per minute” tell us?

Read the reaction accurately

Matter changes; light supplies the energy

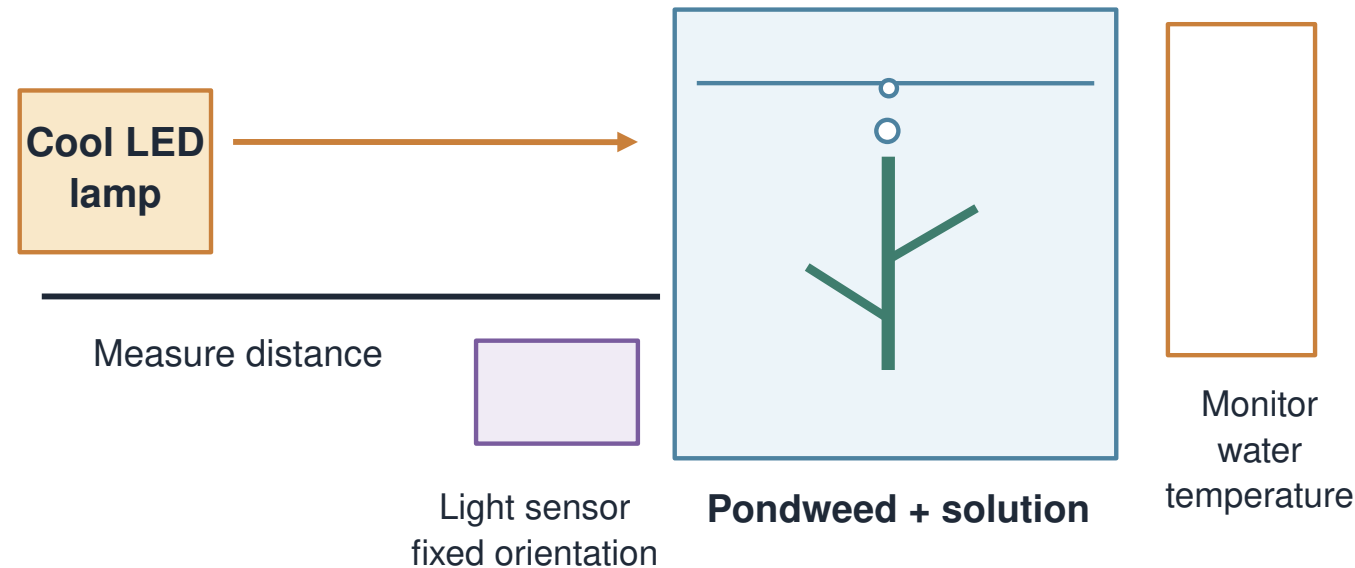


Glucose can build biomass or be used in respiration.
Photosynthesis takes in energy.

Carbon dioxide + water → glucose + oxygen. Light energy is absorbed by chlorophyll. Photosynthesis takes in energy.

I plan the measurement before moving the lamp

Change light; monitor temperature

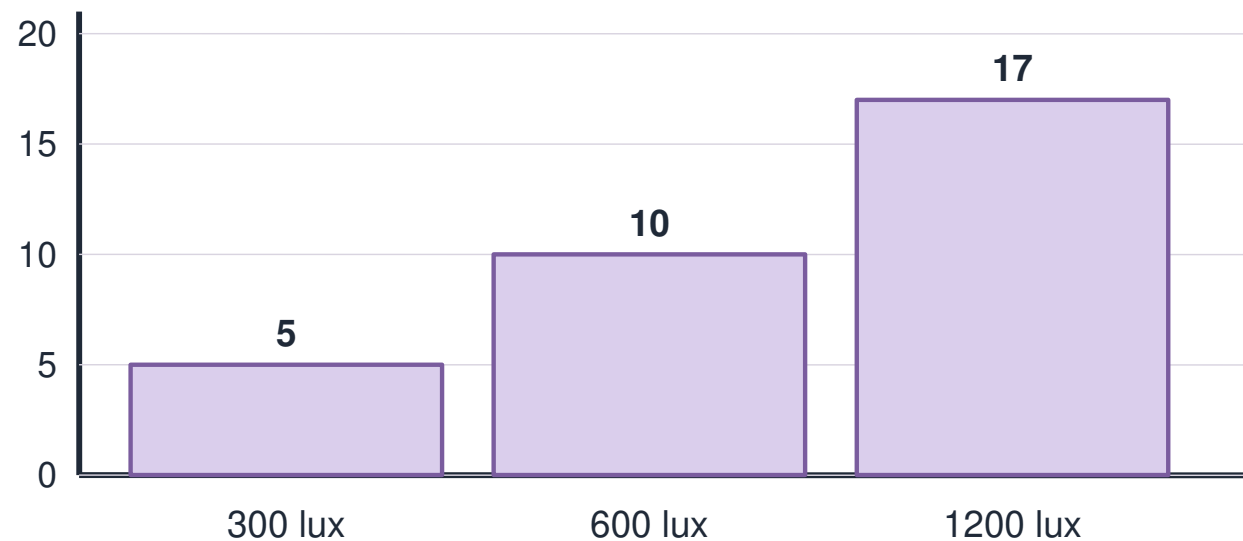


Change the light level and record it. Count bubbles over equal one-minute intervals. Keep the shoot and other conditions comparable.

Work a rate and a mean

Constructed results: compare mean bubble rates

Mean bubbles per minute; light sensor level below



Worked example: 12 bubbles in 2 minutes = 6 per minute. Three one-minute counts 4, 5, 6 have mean 5. Count is a proxy, not an exact oxygen volume.

Investigation team: predict, then calculate

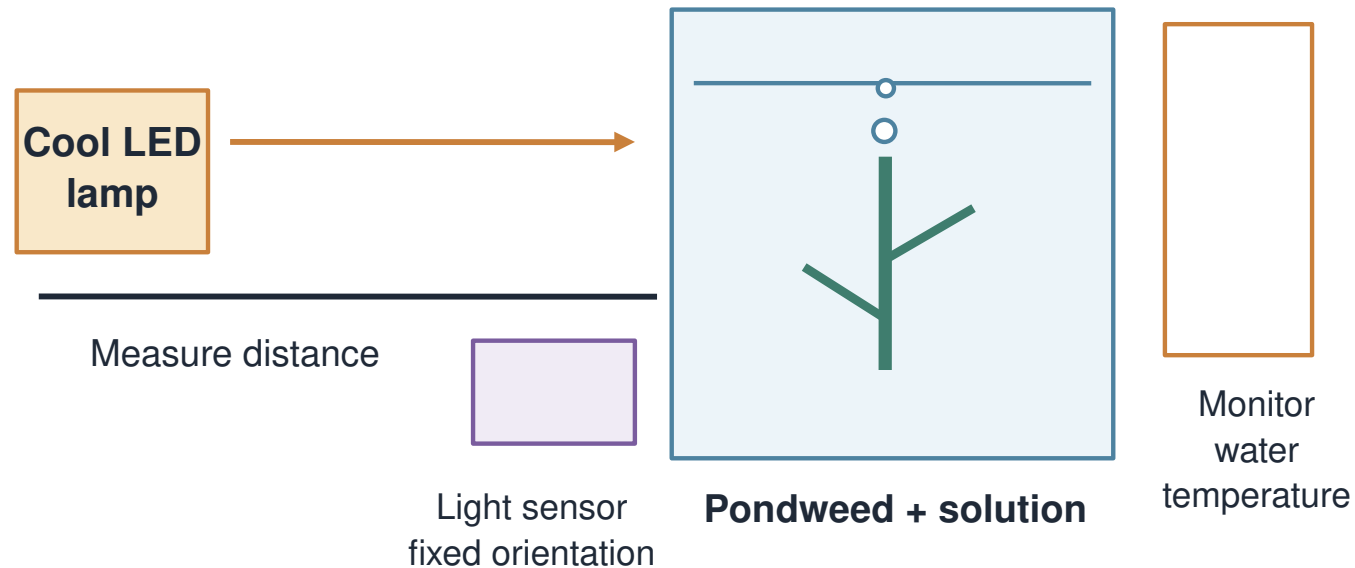
W1. Predict the largest mean rate. Calculate the mean for each light level.

Lamp distance(cm)	Mean sensor(lux)	Three 1-min counts
40	300	4,5,6
30	600	9,10,11
20	1200	16,17,18

Constructed sample data; same shoot/solution and 20°C; counts are bubbles.

Method repair challenge

Change light; monitor temperature



Same shoot, solution, timing and background light.

W2. A group changes the shoot and the lamp position together. Can it isolate the light effect? Repair one step.

Which conclusion earns its claim?

W3. “Within this range, higher measured light gave a higher mean count.”

Does that prove unlimited growth or exact oxygen volume?

A valid scientific conclusion

Choose the statement supported by the method.

- A** Oxygen is made from light as a material.
- B** Bubble count can indicate a pattern, but bubble size limits precision.
- C** Twice the bubbles always means exactly twice the oxygen volume.
- D** Changing the shoot and light together isolates light.

Prepare your investigation brief

Choose your response route.

Include a reaction, method and evidence-based conclusion.

Use the evidence independently

Predict → calculate → explain → improve the method.

Keep constructed results separate from any actual observations.

Audience question: what did you measure?

L1. Show your conclusion.

Audience asks: “Count, volume, growth or photosynthesis itself?”

A strong brief names its limitation

The reaction explains the process.

The method determines what the evidence can show.

Keep matter, energy and evidence distinct

E1. Name the two material reactants.

E2. Calculate 15 bubbles in 3 minutes.

E3. Name one control and one proxy limit.

Next useful step

Reaction • rate • fair method • cautious conclusion.